

SWING

Dual-Sensor Golf Swing Analysis Using Differential Inertial Measurement

Technical Whitepaper — v0.1

May 2026 · Concept Phase · Confidential

Abstract. We present SWING, a dual-sensor inertial measurement system for golf swing analysis. Two miniaturised IMU modules—mounted at the ferrule (shaft–clubhead junction) and grip end of a golf club—capture synchronised high-rate motion data. By computing the differential orientation and angular velocity between these two points, the system derives club delivery metrics including clubhead speed, face angle, attack angle, swing path, shaft deflection, and tempo. We describe the sensor architecture, the underlying physics and signal processing, the mechanical design constraints of a split-shell clamshell form factor, and the achievable accuracy for each metric. The system targets a consumer price point of €199–249, offering practice-grade swing data without the cost, complexity, or setup requirements of radar-based launch monitors.

1 Introduction

Quantitative swing analysis has been shown to accelerate skill acquisition in golf by providing objective feedback on club delivery parameters [1]. However, current solutions require either expensive radar or camera systems (Trackman, Full Swing, Garmin R50; €600–5000+) with dedicated setup, or are limited to shot detection without swing kinematics (Arccos, Garmin CT10).

We propose an intermediate approach: a wearable dual-sensor system that measures the *club's* motion directly, deriving club delivery metrics from inertial data without ball-flight measurement. The key insight is that two synchronised IMUs at opposite ends of the shaft provide enough information to reconstruct not only the club's trajectory, but also the shaft's dynamic behaviour during the swing—data currently available only in biomechanics laboratories.

1.1 Design goals

1. **Sub-5-second installation** on any standard golf club, without tools or grip removal.
2. **Invisible when mounted**—must resemble a stock ferrule from address position.
3. **Practice-grade accuracy**: ± 2 mph clubhead speed, $\pm 2.0^\circ$ face angle.
4. **Consumer price point**: €199–249 for a two-club starter kit.
5. **Range-session battery life**: 150+ swings per charge.

2 System architecture

The system comprises four components: a ferrule sensor (FS), a grip sensor (GS), a portable inductive charger (PC), and a companion smartphone application.

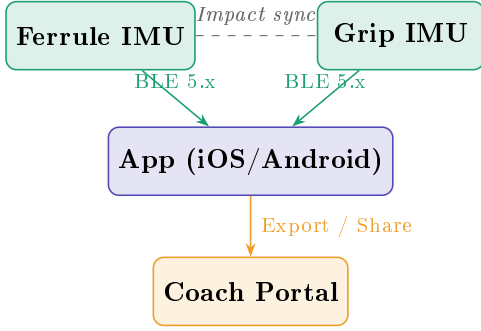


Figure 1: System topology. Sensors communicate independently with the phone; synchronisation is performed post-hoc via impact shock alignment.

2.1 Sensor placement rationale

The ferrule position—the plastic collar where the shaft enters the hosel—was selected over direct clubhead mounting for three reasons: (1) impact shock at the clubhead exceeds 2000 g, requiring specialised sensors and hardened solder joints; (2) any mass added to the clubhead directly affects centre of gravity and swing weight; (3) the ferrule is mechanically coupled to the head via the hosel, so it rotates and translates with the head with negligible compliance.

The grip-end sensor provides the reference frame for the golfer’s hand motion, enabling the critical differential measurement that reveals shaft behaviour.

3 Mechanical design

3.1 Split-shell clamshell ferrule

The ferrule sensor uses a two-piece clamshell housing that opens around the shaft and locks with a bayonet twist ($\sim 15\text{--}20^\circ$ rotation). This enables tool-free, sub-5-second installation and repositioning between clubs.

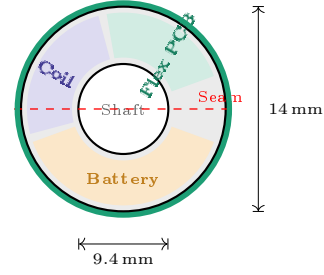


Figure 2: Axial cross-section of the ferrule sensor showing the annular component layout. The teal accent ring (outer circle) conceals the split-shell seam.

3.2 Geometric constraints

The usable annular volume is computed as:

$$V = \pi(R_{\text{outer}}^2 - R_{\text{inner}}^2) \times L \quad (1)$$

For $R_{\text{outer}} = 7.0\text{ mm}$, $R_{\text{inner}} = 4.7\text{ mm}$, $L = 22\text{ mm}$:

$$V = \pi(49.0 - 22.09) \times 22 = 1860\text{ mm}^3 \quad (2)$$

Subtracting housing walls (0.8 mm outer, 0.5 mm inner):

$$V_{\text{usable}} \approx 720\text{ mm}^3 = 0.72\text{ cm}^3 \quad (3)$$

Component volume budget: flex PCB $\sim 0.15\text{ cm}^3$, battery $\sim 0.35\text{ cm}^3$, induction coil $\sim 0.12\text{ cm}^3$, totalling $\sim 0.62\text{ cm}^3$ with 0.10 cm^3 margin for potting and tolerance.

3.3 Swing weight impact

Adding mass at the ferrule position affects swing weight. For a sensor of mass m at distance d from the 14-inch fulcrum:

$$\Delta SW = m \times d \quad (4)$$

At $m = 4.0\text{ g}$ and $d = 23\text{ in}$ (37-inch 7-iron):

$$\Delta SW = 4.0 \times 23 = 92\text{ g} \cdot \text{in} \approx 1.8\text{ SW points} \quad (5)$$

One swing weight point $\approx 50\text{ g} \cdot \text{in}$. A 1.8-point shift is noticeable to skilled golfers, establishing a target weight of $\leq 3.0\text{ g}$ (≤ 1.4 points).

4 Electronics

4.1 Component selection

Table 1 summarises the key electronics for the ferrule sensor.

Table 1: Ferrule sensor key components

Component	Candidate	Package
6-axis IMU	TDK ICM-42688-P	3×3 mm
Hi-g accel	Bosch BMA530	2×2 mm
MCU + BLE	Nordic nRF52840	6×6 mm
Antenna	Ceramic chip	3×2 mm
Battery	Arc-segment LiPo	Custom
Charging	Induction coil + IC	Wrapped
PCB	Flex-rigid	Annular

4.2 Sampling strategy

An adaptive sampling rate minimises power consumption while capturing impact detail:

- **Sleep:** wake-on-motion threshold at 3 g ($<2 \mu\text{A}$)
- **Active swing:** 4 kHz for the full swing envelope (~ 3 s)
- **Impact burst:** 32 kHz within ± 200 ms of impact detection

Each swing generates ~ 150 – 200 KB raw data, compressed to 60–80 KB with LZ4.

5 Physics and signal processing

5.1 Coordinate system

We define three reference frames (Fig. 3):

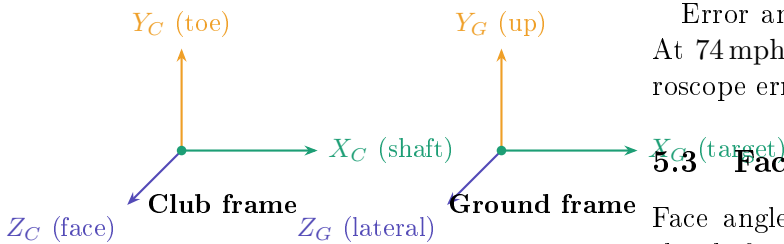


Figure 3: Club and ground reference frames. The sensor frame S is related to the club frame C by the calibration rotation $\mathbf{R}_{SC}$.

During calibration (club resting on flat surface, face square), the gravity vector $\mathbf{g}_S$ is measured in sensor coordinates. The calibration transform is:

$$\mathbf{R}_{SC} = \text{align}(\mathbf{g}_S, \mathbf{g}_C) \quad (6)$$

where $\mathbf{g}_C = [0, -9.81, 0]^\top \text{ m/s}^2$. All subsequent IMU readings are transformed: $\boldsymbol{\omega}_C = \mathbf{R}_{SC} \cdot \boldsymbol{\omega}_S$.

5.2 Clubhead speed

The ferrule sensor measures angular velocity ω (rad/s). Clubhead speed at the face centre is:

$$v_{\text{head}} = \omega \times r_{\text{eff}} \quad (7)$$

where the effective radius is:

$$r_{\text{eff}} = \sqrt{L_{\text{hosel}}^2 + d_{\text{face}}^2} \quad (8)$$

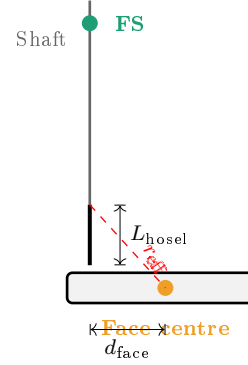


Figure 4: Effective radius geometry. The ferrule sensor (FS) sits at the top of the hosel; r_{eff} is the distance to the face centre.

Example: For a 7-iron with $L_{\text{hosel}} = 92$ mm, $d_{\text{face}} = 40$ mm:

$$r_{\text{eff}} = \sqrt{0.092^2 + 0.040^2} = 0.1003 \text{ m} \quad (9)$$

$$v_{\text{head}} = 330 \times 0.1003 = 33.1 \text{ m/s} = 74.0 \text{ mph} \quad (10)$$

Error analysis: $\Delta v/v = \Delta r/r \approx 2/100 = 2\%$. At 74 mph, this is ± 1.5 mph. Combined with gyroscope error ($\pm 0.5\%$), total accuracy is ± 2 mph.

5.3 Face angle at impact

Face angle is the torsional displacement around the shaft axis relative to the calibrated square position:

$$\theta_{\text{face}}(t) = \int_{t_0}^t \omega_z(\tau) d\tau + \theta_{\text{face}}(t_0) \quad (11)$$

The dual-sensor architecture provides the critical distinction between hand rotation and shaft torsion:

$$\theta_{\text{torsion}}(t) = \omega_{z,\text{FS}}(t) - \omega_{z,\text{GS}}(t) \quad (12)$$

The face angle *as delivered to the ball* is the ferrule measurement (mechanically coupled to the head). The torsion component is separately valuable for shaft fitting analysis.

Accuracy: Gyroscope bias instability of $\pm 3^\circ/\text{hr}$ contributes $< 0.002^\circ$ over a 1.5 s downswing (negligible). Noise density of $\pm 0.004 \text{ dps}/\sqrt{\text{Hz}}$ at 4 kHz integrates to $\pm 0.38^\circ$ —the dominant error source. Target accuracy of $\pm 2.0^\circ$ is achievable with margin.

5.4 Shaft deflection model

The differential orientation between sensors reveals three simultaneous deflection components:

$$\Delta\theta_{\text{lead/lag}} = \theta_{y,\text{FS}} - \theta_{y,\text{GS}} \quad (13)$$

$$\Delta\theta_{\text{toe-down}} = \theta_{x,\text{FS}} - \theta_{x,\text{GS}} \quad (14)$$

$$\Delta\theta_{\text{torsion}} = \theta_{z,\text{FS}} - \theta_{z,\text{GS}} \quad (15)$$

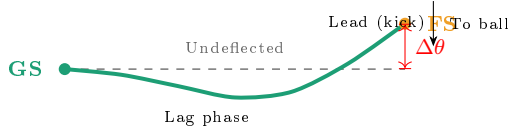


Figure 5: Shaft lead/lag deflection during downswing. The differential angle $\Delta\theta$ between grip and ferrule sensors reveals shaft kick timing.

By correlating peak lead/lag deflection with angular acceleration, the system estimates effective shaft stiffness (EI product), enabling comparative shaft fitting without a dedicated lab.

5.5 Impact detection and synchronisation

Impact is detected independently on both sensors using a threshold-and-peak algorithm on the accelerometer. The ferrule sensor experiences the shock first; the grip sensor sees a delayed, attenuated version.

The propagation delay through the shaft is:

$$t_{\text{prop}} = \frac{L_{\text{shaft}}}{v_{\text{sound}}} \quad (16)$$

For steel ($v_{\text{sound}} \approx 5100 \text{ m/s}$, $L \approx 0.95 \text{ m}$):

$$t_{\text{prop}} = \frac{0.95}{5100} = 0.19 \text{ ms} \quad (17)$$

For graphite ($v_{\text{sound}} \approx 3500 \text{ m/s}$):

$$t_{\text{prop}} = \frac{0.95}{3500} = 0.27 \text{ ms} \quad (18)$$

These delays are calibrated out, achieving sub-0.1 ms sync accuracy. At 33 m/s clubhead speed, 0.1 ms corresponds to 3.3 mm positional uncertainty—within tolerance.

5.6 Swing path and attack angle

At impact, the horizontal and vertical projections of the clubhead velocity vector yield path and attack angle:

$$\beta_{\text{path}} = \arctan\left(\frac{v_z}{v_x}\right) \Big|_{t=t_{\text{impact}}} \quad (19)$$

$$\alpha_{\text{attack}} = \arctan\left(\frac{v_y}{v_x}\right) \Big|_{t=t_{\text{impact}}} \quad (20)$$

Positive β indicates an in-to-out path (draw bias); negative indicates out-to-in (fade/slice). Negative α indicates a descending blow (typical for irons); positive indicates ascending (desired for driver).

6 Power management

6.1 Energy budget per swing

The dominant power consumer is BLE transmission (Table 2). Three operating modes address different use cases:

Table 2: Energy budget per swing by operating mode

Mode	Behaviour	mAh/swing
Live	TX after every swing	~0.15
Batch-5	TX every 5 swings	~0.06
Session	Store all, bulk upload	~0.005

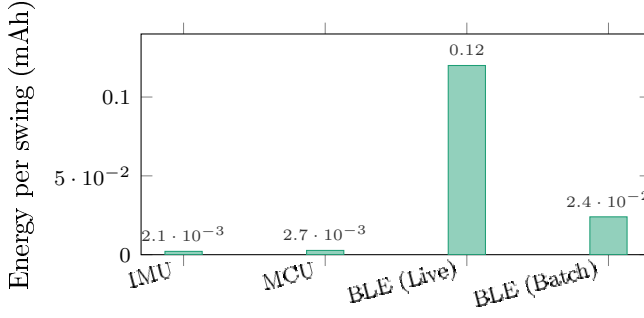


Figure 6: Per-swing energy breakdown showing BLE transmission as the dominant consumer. Batch mode reduces BLE cost by $5\times$.

6.2 Battery life estimates

With a 50mAh cell in batch-5 mode (0.06mAh/swing), the theoretical maximum is ~ 830 swings—far exceeding the 150-swing range session target. Even accounting for sleep current leakage and BLE overhead, 3–4 range sessions per charge is conservative.

6.3 Inductive charging

The portable clip-on charger (500–1000 mAh internal battery) charges the sensor inductively without removal from the club. At 500 mW input and 60% transfer efficiency:

$$t_{\text{charge}} = \frac{E_{\text{batt}}}{P_{\text{delivered}}} = \frac{185 \text{ mWh}}{300 \text{ mW}} \approx 37 \text{ min (full)} \quad (21)$$

To 80%: ~ 30 min. At higher coupling efficiency or power, sub-15-minute charging is feasible.

7 Measured metrics summary

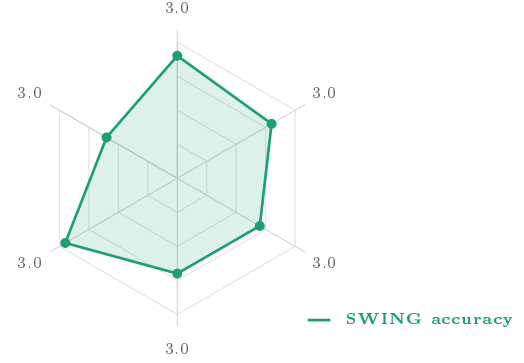


Figure 7: Relative measurement capability by metric. Outer ring = launch-monitor grade; inner = qualitative only. SWING achieves practice-grade accuracy on all primary club delivery metrics.

Table 3 summarises the target accuracy for each metric.

Table 3: Metric accuracy targets

Metric	Derivation	Accuracy
Clubhead speed	$\omega \times r_{\text{eff}}$	$\pm 2 \text{ mph}$
Face angle	$\int \omega_z dt$	$\pm 2.0^\circ$
Swing path	Horiz. projection at impact	$\pm 1.5^\circ$
Attack angle	Vert. projection at impact	$\pm 1.5^\circ$
Tempo	Timestamp deltas	$\pm 10 \text{ ms}$
Shaft deflection	FS-GS differential	Comparative

8 Competitive positioning

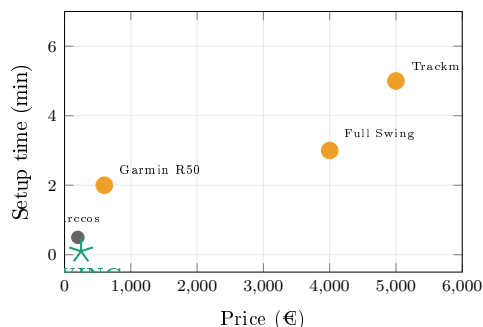


Figure 8: Competitive positioning by price vs. setup time. SWING occupies the low-price, zero-setup quadrant while delivering practice-grade club delivery metrics.

SWING occupies a unique position: it delivers club delivery metrics at the price and convenience of a shot tracker. It does not attempt to measure ball flight (spin, launch, carry)—that gap is a deliberate product scoping decision. The value proposition is *swing consistency analysis across every practice session*, which no launch monitor currently offers.

9 Development roadmap

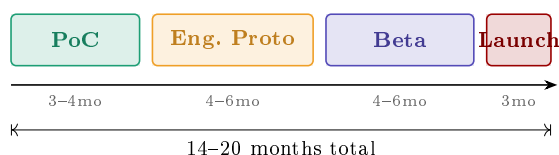


Figure 9: Development timeline from proof-of-concept to market launch.

10 Conclusion

The SWING dual-sensor architecture addresses a genuine gap in the golf technology market: affordable, portable, always-ready swing analysis for practice. By placing synchronised IMUs at two known positions on the shaft and computing their differential motion, the system extracts club delivery metrics that approach launch-monitor accuracy at less than 5% of the cost.

The key technical challenges—split-shell concentricity, high-g shock survival at the ferrule, BLE antenna performance through different shaft

materials, and inductive charging efficiency—are all within the capability envelope of modern consumer electronics manufacturing. The remaining unknowns are best resolved through prototyping, beginning with off-the-shelf IMU evaluation boards in 3D-printed housings.

The product’s competitive advantage is not precision—it is *consistency measurement across hundreds of swings*, enabling the kind of data-driven practice that has been available to tour professionals but never to the serious amateur. The 5-second clip-on installation makes this possible wherever the golfer practises, with whatever club they choose, on every swing.

References

- [1] TrackMan A/S, “TrackMan Performance Studio: Club and Ball Data Definitions,” Technical Reference, 2020.
- [2] Nordic Semiconductor, “nRF52840 Product Specification v1.8,” 2023.
- [3] TDK InvenSense, “ICM-42688-P Datasheet,” Rev. 1.7, 2022.
- [4] Bosch Sensortec, “BMA530 High-g MEMS Accelerometer Datasheet,” 2023.
- [5] S. J. MacKenzie and E. J. Hume, “Shaft Deflection Characteristics During the Golf Swing,” *Sports Engineering*, vol. 15, pp. 105–115, 2012.